

Cost Drivers of Endothelial Keratoplasty

A Time-Driven Activity-Based Costing Analysis

Jenna K. Goldstein, BA,¹ David S. Portney, MD,² Rebecca Kirby, MBA,³ Angela Verkade, MD,² Shahzad I. Mian, MD²

Purpose: To determine cost drivers of endothelial keratoplasty (EK) through evaluation of surgical costs and procedure length based on type of EK, use of preloaded grafts, and performance of simultaneous cataract surgery.

Design: This study was an economic analysis of EKs at a single academic institution using time-driven activity-based costing (TDABC) methodology.

Participants: Endothelial keratoplasty surgical cases, including Descemet membrane endothelial keratoplasty (DMEK) and Descemet stripping automated endothelial keratoplasty (DSAEK), at the University of Michigan Kellogg Eye Center from 2016 to 2018 were included in the analysis.

Methods: Data and inputs were obtained via the electronic health record (EHR) and from prior literature. Simultaneous cataract surgeries were included and separately categorized for analysis. Endothelial keratoplasty expenses were determined with TDABC, a method for cost calculation that incorporates the time that key resources are used and each resource's associated cost rate.

Main Outcome Measures: Main outcome measures included surgery length (in minutes) and day-of-surgery costs.

Results: There were 559 EKs included: 355 DMEKs and 204 DSAEKs. Fewer DSAEKs had simultaneous cataract extraction (47; 23%) than DMEK (169; 48%). Of the DMEKs, 196 (55%) used preloaded corneal grafts. Descemet membrane endothelial keratoplasty cost \$392.31 less (95% confidence interval, \$251.05–\$533.57; $P < 0.0001$) than DSAEK and required 16.94 fewer minutes (14.16–19.73; $P < 0.0001$). Descemet membrane endothelial keratoplasty cases that used preloaded corneal grafts cost \$460.19 less (\$316.23–\$604.14; $P < 0.0001$) and were 14.16 minutes shorter (11.39–16.93; $P < 0.0001$). In multivariate regression, preloaded graft use saved \$457.19, DMEK (compared with DSAEK) saved \$349.97, and simultaneous cataract surgery added \$855.17 in day-of-surgery costs.

Conclusions: Cost analysis of TDABC identified a day-of-surgery cost and surgical time reduction associated with the use of preloaded grafts for DMEK, DMEK compared with DSAEK, and isolated EK compared with EK combined with cataract surgery. This study provides an improved understanding of surgical cost drivers and margin incentivization, which may explain trends and indirectly influence patient care decisions in cornea surgery practices.

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Endothelial keratoplasty (EK) procedures have become the preferred method of corneal transplantation in the United States, accounting for 61% of total corneal transplants in 2021.¹ Moreover, the Eye Bank Association of America reported an increase of 15.3% in EKs in 2021, with comparable amounts of Descemet membrane endothelial keratoplasty (DMEK) and Descemet stripping automated endothelial keratoplasty (DSAEK) procedures. This is mainly due to increasing frequencies of DMEK procedures.¹ Descemet membrane endothelial keratoplasty is viewed as superior to DSAEK, with better postoperative visual acuity, fewer rejection episodes, and a faster recovery time.^{2–4} Outside of higher rebubbling rates after

DMEK, complication rates are comparable for both procedures, and the use of precut, prestripped donor corneal tissues for DMEK (pDMEK) has helped to reduce its technical challenges.⁵

As the frequency of EKs increase, so do associated costs, with total costs approaching \$340 million in 2019,⁶ necessitating studies to investigate the costs of various EK types and surgical approaches. A recent review concluded that surgical time could be reduced with preloaded grafts, thereby reducing net surgical costs, although the cited studies had limited sample sizes and did not directly assess for changes in operation times.^{5,7,8} Böhm et al³ compared pDMEK and nonpreloaded DMEK (npDMEK) in a

retrospective cost-effectiveness analysis, concluding that pDMEK was less costly from both a healthcare and societal perspective, although the costing methodology applied simplistic assumptions for surgical resource costs and sample size was limited. Gibbons et al⁹ examined surgical costs for DSAEK and DMEK using a simulated patient undergoing EK. They concluded that DMEK was both less costly and had improved outcomes; however, their analysis did not evaluate surgical costs directly and instead assessed third-party payer costs.

Although previous studies have evaluated costs of EK at a high level, these studies largely focused on societal costs, applying top-down methods with the use of payer reimbursement rates and quality-adjusted life-years to estimate costs.^{3,9} Such methods may be less accurate when measuring direct process costs.¹⁰ To more accurately calculate these costs, we sought to use time-driven activity-based costing (TDABC), a bottom-up costing methodology popularized by Kaplan and Anderson,¹¹ to assess for differences in surgical costs and surgery length based on type of EK, use of preloaded grafts, and performance of simultaneous cataract surgery with EK.

Methods

Study Population and Data Collection

This study was an economic analysis of EKs at a single academic institution, allowing for “provider costs” to be allocated to a single entity, including both the surgeon and the facility. Endothelial keratoplasty surgeries (Current Procedural Terminology [CPT] code 65756) performed at the University of Michigan Kellogg Eye Center from 2016 to 2018 were included (both DMEK and DSAEK). The years in question were included because the introduction of pDMEK occurred in 2017 at our institution and allowed for a mix of npDMEK and pDMEK. Simultaneous cataract surgery, both simple (CPT 66984) and complex (CPT 66982), were included and separately categorized for analysis. Data and inputs were obtained via deidentified electronic health record (EHR) data (Epic Systems). This study was deemed exempt by the University of Michigan institutional review board.

Time-Driven Activity-Based Costing

Time-driven activity-based costing, developed by Kaplan and Anderson,¹¹ calculates costs by determining the time that key resources are used and each resource’s associated cost rate (see Supplementary Appendix, available at www.aaojournal.org). The process paralleled similar work done at the same institution for cataract surgery.¹² We evaluated EK, with the value chain limited to the day of operation. This decision was made for 2 primary reasons: the study’s aim was to understand surgical cost drivers, and there was a lack of availability of large-scale data for clinic visits.

Process Flow

A flow diagram was created on the basis of case observation. The processes included preoperative, perioperative, and postoperative periods. Time stamp data were inputted into the EHR by nursing staff during each step of the process, which was recognized during case observations. The costs associated with each arena included (1) preoperative nursing care; (2) perioperative surgeon, anesthesia,

operating room (OR) time, OR personnel, materials; and (3) postoperative nursing care (Fig 1).

Time Estimates

Time data were supplied by the EHR and acquired via an internal query program. Surgery-related data were downloaded to Microsoft Excel (Microsoft Corporation) and included the procedure, type of graft, date, surgeon, and time stamps for all steps associated with the event, including preoperative assignment, OR start, anesthesia start, surgery start, surgery end, anesthesia end, OR end, and postoperative end times. Time calculations for each resource were determined on the basis of differences between various time stamps. Incomplete timestamp data were examined and manually filled from the EHR. Preoperative care minutes were calculated by the difference between preoperative assignment time and OR start time. Anesthesia minutes were calculated by the difference between anesthesia start and anesthesia end times. The OR minutes were calculated by the difference between OR start and end times. Surgery minutes were calculated by using the difference between surgery start and surgery end times, excluding preparation and draping time due to lack of accurate time stamp data. Therefore, total surgery time is underestimated. Postoperative care unit minutes were calculated by the difference between anesthesia end and postoperative end times.

Labor and Space Costs

Our analysis of EKs was restricted to a single academic health center that owns the OR facilities and instruments. Additionally, the center employs all surgeons and other personnel (e.g., circulating nurses, scrub technicians). Surgeon costs were based on department estimates for salary in 2021, bonuses, benefits, and amount of time worked. Anesthesia costs were calculated with a top-down approach using total allocated anesthesia costs and total OR minutes in 2021. Associated OR costs, including personnel, were estimated on the basis of a previous study¹³ and adjusted for inflation to the year 2021.¹⁴ Likewise, preoperative and postoperative costs were estimated from a previous study and adjusted for inflation.¹⁵ Additional information on capacity cost rates can be found in the Supplementary Appendix (available at www.aaojournal.org).

Supply Costs

Supplies and tools used for each surgery were listed in the EHR as the cost incurred by the institution. The total nongraft costs (e.g., surgical tools) in addition to the cost of implants and corneal grafts were specifically documented for each case. Of note, this excluded medication costs, including staining medications such as Trypan blue. Because of its relative expense and the differential use in pDMEK and npDMEK, Trypan blue costs were separately added to the supply costs for all npDMEK cases based on an internally quoted figure (\$143.80).

Outcomes and Analysis

The main outcome measures were day-of-surgery costs and surgery length (in minutes). Comparisons were made between DMEK and DSAEK, pDMEK and npDMEK, and EKs with and without simultaneous cataract surgery. Student *t* tests were used to compare differences, assuming a *P* value of < 0.05 for statistical significance. Additionally, a multivariate linear regression of the statistically significant variables was conducted to assess for independent effect on day-of-surgery costs and surgery length. Stata (StataCorp LLC) was used for analysis.

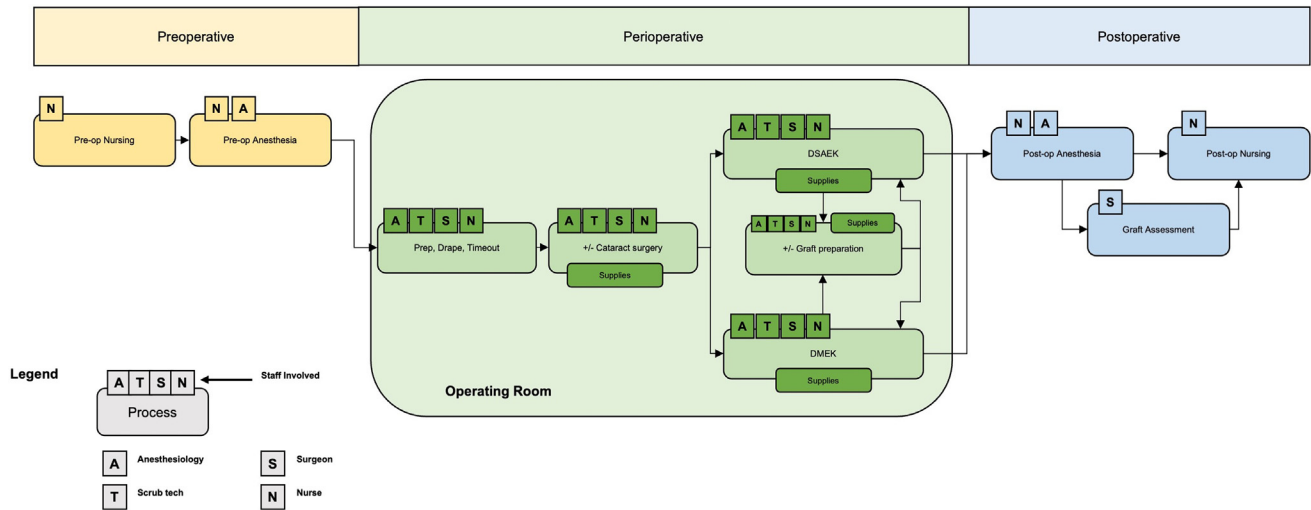


Figure 1. Endothelial keratoplasty (EK) surgery process flow diagram. Both types of EK share similar preoperative and postoperative processes. The operating room (OR) process may be differentiated by the presence of coexisting cataract surgery and whether the surgeon is required to prepare the graft, which can be avoided if using a preloaded graft (in DMEK). DMEK = Descemet membrane endothelial keratoplasty; DSAEK = Descemet stripping automated endothelial keratoplasty.

Results

From 2016 to 2018, a total of 559 EKs were included, of which there were 355 DMEK and 204 DSAEK. Nine procedures (1 DMEK; 8 DSAEKs) were excluded because of the combination with other surgeries (not cataract surgery) or inadequate data. Fewer DSAEKs had simultaneous cataract extraction (47; 23%) than DMEK (169; 48%). Of the DMEKs, 196 (55%) used preloaded corneal grafts.

Average surgery times for each procedure and subset of procedure are shown in Table 1. The DMEK cases were shorter than DSAEK cases (37.85 vs. 54.79 minutes, a difference of 16.94 minutes; $P < 0.001$). Cases with cataract surgery added 11.04 minutes ($P < 0.001$) for DMEK and 14.50 minutes ($P < 0.001$) for DSAEK. Performance of pDMEK saved 14.16 minutes ($P < 0.001$) compared with npDMEK. Procedure time was less for pDMEK compared with npDMEK when evaluating simultaneous cataract surgery (reduction of 16.52 minutes; $P < 0.001$) and

DMEK without cataract surgery (reduction of 11.94 minutes; $P < 0.001$).

Average day-of-surgery costs for each procedure and subset of procedure are shown in Table 2 and Figure 2. The DMEK cases were less costly than the DSAEK cases (\$7443.18 vs. \$7835.49, difference of \$392.31; $P < 0.001$). Cases with cataract surgery added \$750.75 ($P < 0.001$) for DMEK and \$1114.69 ($P < 0.001$) for DSAEK. The cost was \$460.19 ($P < 0.001$) less for pDMEK than for npDMEK. Furthermore, pDMEK was found to be cost-saving for day-of-surgery compared with npDMEK when evaluating simultaneous cataract surgery (reduction of \$574.39; $P < 0.001$) and simple DMEK (reduction of \$351.26; $P = 0.0002$).

The results of multivariable regression analysis are shown in Table 3. Descemet membrane endothelial keratoplasty reduced surgery length by 12.17 minutes compared with DSAEK, pDMEK reduced surgery length by 14.12 minutes compared with npDMEK, and performing a simultaneous cataract surgery added 12.30 minutes. For day-of-surgery costs, DMEK cost \$349.97 less

Table 1. Average Surgery Length (Minutes) Grouped by Type of Endothelial Keratoplasty, Coexisting Cataract Extraction, and Use of a Preloaded Graft

Group	Count	Average Time	SD	Average Difference	95% CI	P Value
DMEK	355	37.85	14.96			
DSAEK	204	54.79	18.04	16.94	14.16 to 19.73	< 0.0001
DMEK with cataract	169	43.63	13.08			
DMEK without cataract	186	32.59	13.64	-11.04	-13.95 to -8.13	< 0.0001
DSAEK with cataract	47	66.72	14.72			
DSAEK without cataract	157	52.22	17.43	-14.50	-21.03 to -9.98	< 0.0001
DMEK with preloaded graft	196	31.51	13.44			
DMEK without preloaded graft	159	45.67	12.92	14.16	11.39 to 16.93	< 0.0001
DMEK with cataract with preloaded graft	93	36.20	8.05			
DMEK with cataract without preloaded graft	76	52.72	12.31	16.52	13.41 to 19.63	< 0.0001
DMEK without cataract with preloaded graft	103	27.26	15.77			
DMEK without cataract without preloaded graft	83	39.20	9.72	11.94	8.04 to 15.85	< 0.0001

CI = confidence interval; DMEK = Descemet membrane endothelial keratoplasty; DSAEK = Descemet stripping automated endothelial keratoplasty; SD = standard deviation.

Table 2. Average Day-of-Surgery Costs Grouped by Type of Endothelial Keratoplasty, Coexisting Cataract Extraction, and Use of a Preloaded Graft

Group	Count	Average Cost	SD	Average Difference	95% CI	P Value
DMEK	355	\$7443.18	\$722.18			
DSAEK	204	\$7835.49	\$963.83	\$392.31	\$251.05 to \$533.57	< 0.0001
DMEK with cataract	169	\$7836.53	\$584.24			
DMEK without cataract	186	\$7085.78	\$646.71	−\$750.75	−\$879.86 to −\$621.63	< 0.0001
DSAEK with cataract	47	\$8693.36	\$735.98			
DSAEK without cataract	157	\$7578.67	\$872.37	−\$1114.69	−\$1391.14 to −\$838.22	< 0.0001
DMEK with preloaded graft	196	\$7237.07	\$678.63			
DMEK without preloaded graft	159	\$7697.25	\$694.59	\$460.19	\$316.23 to \$604.14	< 0.0001
DMEK with cataract with preloaded graft	93	\$7578.22	\$399.26			
DMEK with cataract without preloaded graft	76	\$8152.61	\$620.54	\$574.39	\$418.50 to \$730.28	< 0.0001
DMEK without cataract with preloaded graft	103	\$6929.03	\$731.25			
DMEK without cataract without preloaded graft	83	\$7280.29	\$457.46	\$351.26	\$169.60 to \$532.93	0.0002

CI = confidence interval; DMEK = Descemet membrane endothelial keratoplasty; DSAEK = Descemet stripping automated endothelial keratoplasty; SD = standard deviation.

than DSAEK, pDMEK cost \$457.19 less than npDMEK, and performance of simultaneous cataract surgery added \$855.17 when combined with EK.

Discussion

In this study, we evaluated drivers of EK surgery length and day-of-surgery costs using time-driven activity-based costing. Specifically, there were increased surgical times for DSAEK when compared with DMEK, with use of npDMEK grafts, and for simultaneous cataract removal. The findings for surgical time mirrored those for day-of-surgery

costs. To parse for correlated factors, a multivariate analysis was performed, illustrating that DSAEK is more costly than DMEK, pDMEK saves both time and costs compared with npDMEK, and performing cataract surgery in addition to EK adds significant cost and surgical time.

The current study builds on previous findings, strengthening claims that DMEK is cost-saving compared with DSAEK.⁹ In our analysis, the average DMEK case cost approximately \$500 less and required approximately 17 fewer minutes than DSAEK. Furthermore, in our regression analysis, DMEK saved \$349.97 and 12.17 minutes of surgery. Gibbons et al⁹ found analogous day-of-surgery costs for simple DMEK to be approximately \$100 less

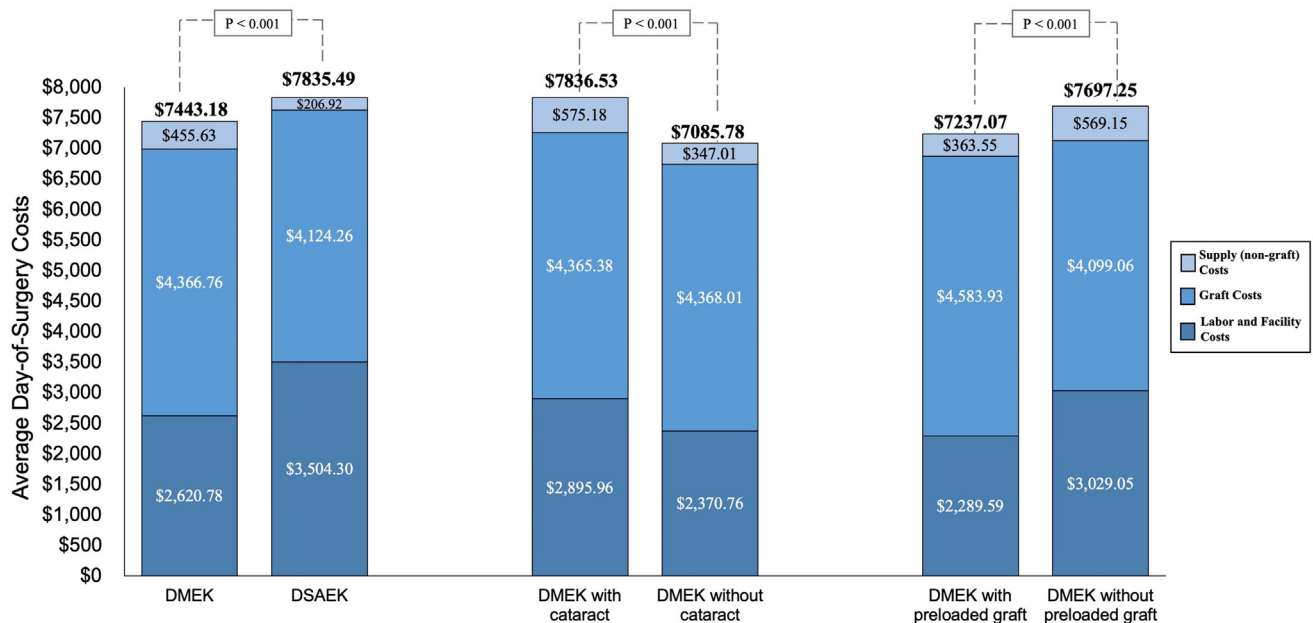


Figure 2. Average day-of-surgery costs grouped by type of endothelial keratoplasty (EK), concurrent cataract extraction, and use of a preloaded graft. DMEK = Descemet membrane endothelial keratoplasty; DSAEK = Descemet stripping automated endothelial keratoplasty; TDABC = time-driven activity-based costing.

Table 3. Multivariate Regression Analysis for Drivers of Endothelial Corneal Transplant Length and Costs

	Coefficient	95% CI
Day-of-surgery costs (\$)		
Base case cost	\$7638.46	\$7539.69 to \$7737.23
DMEK (instead of DSAEK)	−\$349.97	−\$496.24 to −\$203.71
Coexisting cataract extraction	\$855.17	\$734.00 to \$976.34
Use of a preloaded graft	−\$457.19	−\$601.62 to −\$312.77
Surgery length (min)		
Base case cost	51.96	49.95 to 53.97
DMEK (instead of DSAEK)	−12.17	−15.15 to −9.20
Coexisting cataract extraction	12.30	9.84 to 14.76
Use of a preloaded graft	−14.12	(−17.05 to −11.18)

CI = confidence interval; DMEK = Descemet membrane endothelial keratoplasty; DSAEK = Descemet stripping automated endothelial keratoplasty.

costly than DSAEK. However, in this analysis, they did not account for increased facility costs or surgeon costs associated with increased surgery length and, therefore, likely underestimated the cost-savings associated with DMEK. Additionally, their simulated analysis accounted for the postoperative courses for both procedures, assuming DMEK to have slightly increased rates of rebubbling but better final visual acuity. Despite this complication, DMEK resulted in both superior visual outcomes and a more cost-effective procedure. Our analysis provides more concrete data surrounding the cost-savings of DMEK.

A second key finding in our study was the notable cost-savings associated with preloaded corneal grafts. In our multivariate analysis, pDMEK was found to be both cost-effective and time-saving, costing \$457.19 less and requiring 14.12 fewer minutes of surgery. At our institution, preloaded grafts were \$500 more costly than nonpreloaded grafts, although some cost increase was offset by the reduced need for Trypan blue (\$143.80). With the per-minute operative expense of the surgeon, anesthesiologist, and OR (\$43.96 per minute), estimated costs break even with approximately 8 minutes of reduced surgical time. Therefore, pDMEK generally exceeds this threshold and provides cost-saving. Prior work has highlighted the potential for cost savings with pDMEK grafts. Gibbons et al⁹ also compared pDMEK with npDMEK and found pDMEK to be \$97.58 less, although this analysis was limited as described earlier, not accounting for the OR or surgeon costs. Böhm et al³ compared pDMEK and npDMEK in a retrospective analysis, finding pDMEK to be approximately \$1400 less costly. Of note, their analysis did account for the cost of increased surgery time, finding that npDMEK was approximately 1.56 times longer and therefore multiplying all direct costs by this factor. These prior studies were limited in both sample size and costing methodology, and our TDABC analysis provides more detail on the cost differences of these surgical techniques. Given that outcomes for pDMEK are similar to those of npDMEK^{3,16} with less risk of damage to donor tissues⁵ and at a lower cost, it is reasonable to consider pDMEK as the standard of care. This would allow for

standardization of tissues, simplification of surgery, and better allocation of resources, while also shortening surgery time and reducing provider costs.

Furthermore, we found that EK with simultaneous cataract surgery lengthened surgical time and was more costly than EK alone. Multivariate analysis revealed concurrent cataract surgery for all EKs to be \$855.17 costlier, mostly due to an additional 12.30 minutes of surgery. Prior cost analyses identified a wide variety of cost benefits when combining phacoemulsification with vitrectomy, such as one-time use fees of medications and supplies, a reduction in transportation costs, and an overall reduced total payer fee.^{17,18} Although vitrectomy differs from EK, many of the same financial benefits of combined surgery are likely applicable to EK. Our study adds to existing literature by quantifying the incremental cost of cataract extraction during EK. Although day-of-costs are higher for concurrent cataract procedures, the incremental cost of \$855.17 is significantly cheaper than performing an independent simple cataract surgery, because a recent study by Portney et al¹² at the same institution estimated day of costs to total approximately \$1900 for simple cataract surgery. Moreover, a notable limitation of our analysis was the exclusion of preoperative and postoperative care, limiting our ability to calculate total costs. Taking this into consideration, potential cost-savings from combining surgeries may be even greater than hypothesized because clinical care can be combined. Although phakic DMEK may help to maintain lens accommodative function in younger patients and improve keratometry measurements necessary for optimal determination of lens power,¹⁹ inclusion of cataract surgery during an EK procedure eliminates the incidence of further endothelial damage from sequential cataract surgery, while yielding faster visual rehabilitation and comparable clinical outcomes to lone EKs.^{20,21}

The exploration of cost drivers would be incomplete without also comparing such costs with revenue, allowing for an evaluation of net margin. At our institution, the costs of EK grafts are passed through to the patient and payer. Therefore, the increased implant cost of preloaded grafts is not directly absorbed by the provider, further increasing the cost incentive to use preloaded grafts, which then have an approximately \$957 positive impact on net margin compared with nonpreloaded grafts (\$500 in pass-through graft costs in addition to the \$457.19 identified in the regression analysis). This example highlights both a cost and quality improvement for preloaded DMEK, in which financial and quality incentives are aligned. However, although performing cataract surgery during an EK procedure leads to lower total costs and is thought to be superior in quality and safety to performing separate surgeries, it may not improve net margin for surgeons. With combined or bilateral surgeries, the lower reimbursed second surgery (cataract surgery) is reimbursed at a 50% rate by Medicare,²² yielding a Medicare professional payment of approximately \$276 in the locality in which this analysis took place,²³ well below the incremental cost of simultaneous cataract surgery in our study. Therefore, cornea surgeons may be disincentivized from combined cases, a phenomenon similar to that seen with same-day

bilateral cataract surgery.^{22,24,25} In the case of DMEK, however, combined cases are superior when considering reduced incremental costs, endothelial damage, and anesthesia risks compared with successive surgeries. Such disincentivizing may have negative downstream effects, and increasing professional fee reimbursement, or at least not reducing it by 50% for combined cases, may be reasonable. It is important to note that our analysis is unable to adequately address the true total costs of these surgical bundles due to the exclusion of postoperative visits, underestimation of surgeon time from prepping and draping, and exclusion of intraoperative medications (i.e., basic salt solution). Thus, further work is necessary to better understand gross margin and necessary reimbursement changes.

Study Limitations

Our study has several important limitations. Given the setting of our data at a large academic center, results may not be as applicable to private practice ophthalmologists with different procedural methods and contract agreements. A key contributor to cost differences across facilities is variation in supplies and materials expenses,²⁶ and the values used in this study may differ greatly from those at other surgical centers. Although this is true, focusing on a single institution allows for more detailed analysis of costs and resource use, which can still be informative to other organizations looking for cost-saving strategies. Second, the institution uses a limited number of cornea surgeons, including fellows with little EK experience, thereby introducing greater variation in surgical expertise and procedure times. Because of this, we may be overestimating surgery costs and length, especially in the more technically challenging DMEK.^{2,9} Despite this discrepancy, we found DMEK to be significantly cheaper and quicker than DSAEK. Furthermore, we provided detailed information

about our calculations, allowing for transparency and replication of this study's methods with a larger sample of surgeons. Another limitation is that pDMEK did not become a standard of practice until 2017, whereas our analysis of npDMEK includes procedures from 2016, possibly impacting cost evaluations. The impact of procedure year, however, is negligible because all cost inputs were adjusted for inflation to the year 2021 and similar amounts of pDMEK and npDMEK were performed overall (55% pDMEK). Furthermore, because this was a retrospective study, we were unable to control for several variables that would likely impact costs, including surgery complexity. It is reasonable that some of the increase in surgery length and costs, for example, with DSAEK versus DMEK, is a result of increased case complexity as indications for DSAEK and DMEK have evolved with increased adaptation of DMEK. Still, TDABC is a well-established method that provides a more accurate picture of direct costs compared with other calculations, and our analysis adds to an overall understanding of the impact of a variety of factors for EK.

Conclusions

Our study used TDABC to identify the day-of-surgery cost drivers for EK, finding a reduction in surgical time and costs associated with DMEK compared with DSAEK and savings associated with the use of preloaded grafts. We also quantified the impact of simultaneous cataract surgery on the cost of surgery, noting that incremental surgical costs may exceed incremental reimbursement. These findings and the relative reimbursements of these surgeries may explain cornea surgeon practice patterns and ultimately impact patient care.

Footnotes and Disclosures

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¹ University of Michigan Medical School, Ann Arbor, Michigan.

² Kellogg Eye Center, Department of Ophthalmology and Visual Sciences, University of Michigan, Ann Arbor, Michigan.

³ Kellogg School of Management at Northwestern University, Evanston, Illinois.

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Author Contributions:

Conception and design: Goldstein, Portney

Data collection: Goldstein, Portney

Analysis and interpretation: Goldstein, Portney, Kirby

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Abbreviations and Acronyms:

CPT = Current Procedural Terminology; **DMEK** = Descemet membrane endothelial keratoplasty; **DSAEK** = Descemet stripping automated endothelial keratoplasty; **EHR** = electronic health record; **EK** = endothelial keratoplasty; **npDMEK** = nonpreloaded Descemet membrane endothelial keratoplasty; **OR** = operating room; **pDMEK** = prestripped donor corneal tissues for DMEK; **TDABC** = time-driven activity-based costing.

Keywords:

Cornea, Descemet membrane endothelial keratoplasty, Descemet stripping automated endothelial keratoplasty, Preloaded grafts, Time-driven activity-based costing.

Correspondence:

Shahzad I. Mian, MD, Kellogg Eye Center, University of Michigan, 1000 Wall Street, Ann Arbor, MI 48105. E-mail: smian@med.umich.edu.

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